

KHALED YOUNES

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SKILLS SUMMARY

Programming	Python, MATLAB
Software	Gaussian, LAMMPS, Ansys CFX/Fluent/AIM, OpenFOAM, SolidWorks, AutoCAD
Concepts	Data Analysis, Computational Fluid Dynamics, Analytical & Numerical Modeling, Monte Carlo Methods, Bayesian Optimization
Coursework	Plasma Dynamics, Numerical Methods, Fluid Mechanics, Quantum Mechanics, Statistical Mechanics, Differential Equations, Turbulence, Machine Learning

EDUCATION

Stanford University	CA, USA
Ph.D. in Mechanical Engineering	09/2021 – 08/2026 (Expected)
<i>Thesis: Characterizing the Picosecond Dynamics of Liquid Water Subject to Hard X-ray Irradiation</i>	
University of Waterloo	ON, Canada
M.A Sc. in Mechanical Engineering	09/2019 – 08/2021
<i>Thesis: Velocity Scaling of High-Speed Turbulent Boundary Layer Flows with Wall Heat Transfer</i>	
<i>Honors: Finalist for the Alumni & Governor General's Gold Medals</i>	
B.A Sc. in Mechanical Engineering	09/2012 – 06/2017
<i>Honors: With Distinction, Dean's Honours List</i>	

RELEVANT EXPERIENCE

Graduate Research Assistant <i>Stanford University</i>	09/2021 – Present
• Conducted hard X-ray pump hard X-ray probe experiments on liquid water radiolysis at the SLAC National Laboratory and the SPring-8 Angstrom Compact LAsEr facility (SACLA). • Post-processed large data sets (>15 TB) to extract the ultrafast ionization dynamics and quantify their impact on the formation of radiation track structures. • Performed large-scale atomistic simulations using LAMMPS, paired with a Monte-Carlo framework, to model the multi-step ionization cascade following core ionization in liquid water. • Developed a wave equation model to validate the numerical simulations and provide a comprehensive understanding of the radiation effects. • Summarized and presented the findings in technical reports, international conferences, and departmental presentations, effectively communicating the concepts to specialized and broad audiences. • Implemented Bayesian optimization to achieve a 2x increase in state sampling of complex phase spaces for supercritical fluid applications.	
Research Assistant <i>University of Waterloo</i>	07/2017 – 07/2021
• Devised a wall model based on energy conservation for velocity scaling in hypersonic flows with heat transfer. • Developed low-order analytical models in Python to compute the noise generated in gas turbine nozzles and study the fluidic thrust shock-vectoring response of rocket engines. • Conducted high-fidelity detached eddy simulations in SU² to quantify the shock-vectoring performance of hybrid rocket engines under various injection flow rates. • Performed multi-phase, chemically active numerical simulations in OpenFOAM, modeling the thermal decomposition of H₂O₂ for industrial NO_x emissions reduction.	

Aerodynamic Analysis Lead | *H2Drones*

05/2016 – 05/2017

- Performed 3-D CFD computations using Ansys AIM & CFX, evaluating the aerodynamic performance of the vehicle during take-off, landing, and cruise.
- Created 2-D dynamic flight simulation models in MATLAB to study the vehicle response and stability.
- Carried out model and full-scale wind tunnel tests, validating the numerical tools developed.

PREVIOUS INTERNSHIPS

Manufacturing Engineer | *Toyota Motor Manufacturing Canada*

09/2016 – 12/2016

- Applied the Toyota Business Practice methodology to create a standard work procedure for detecting and reporting quality defects in the press shop.
- Resolved downtime causes by running line-side trials, presenting results to upper management and sharing findings with other plants.
- Liaised with steel suppliers and external contractors to ensure the delivery of defect-free blanking coils.
- Trained production team members to improve existing processes and decrease their cycle time.

Mechanical Engineer | *Christie Digital*

01/2016 – 04/2016

- Analyzed and developed the thermal systems of the next-generation Christie Spyder video processor.
- Carried out parametric studies on the fan cooling unit to optimize the air flow across the electronics subsystems of the processor.
- Conducted component and system sound level tests, striving for minimal noise levels through continuous improvements to the fan selection process.

Mechanical Designer | *Inertia Engineering*

05/2015 – 08/2015

- Led the mechanical design of an assisted rescue system for helicopter pilots using SolidWorks.
- Executed design of experiments to determine cable pull out effort in the assisted rescue system.

Materials Engineering | *Linamar Corporation*

09/2014 – 12/2014

- Realized a \$25,000 annual saving by designing a multi-purpose assembly cart, eliminating the need of a forklift for the transportation of 36 parts.
- Increased warehouse storage space by 65% by designing a new racking system in AutoCAD.
- Performed quality checks on more than 75 parts spanning all 4 assembly lines.

TEACHING EXPERIENCE

Graduate Teaching Assistant | *Stanford Mechanical Engineering*

01/2024 – 04/2026

- Delivered lectures and bi-quarterly review sessions to 60+ students on advanced thermodynamic concepts, as part of ME 30: Engineering Thermodynamics and ME 357: Gas Turbine Analysis.
- Prepared and graded midterm and final examinations, regularly monitoring student progress and ensuring fair assessment.
- Held weekly help sessions to provide one-on-one support on course material.

Graduate Teaching Assistant | *Waterloo Faculty of Engineering*

01/2020 – 05/2021

- Responsible for running tutorial sessions and marking exams for 85+ third-year mechanical engineering students, as part of ME 353: Heat Transfer and ME 351: Fluids Mechanics I.

In-House Residence Tutor | *Claudette Millar Hall Residence*

09/2017 – 04/2018

- Tutored 10 first-year engineering students Calculus on a weekly basis.